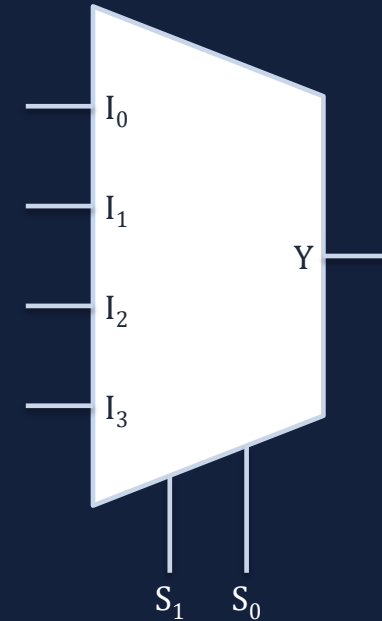


Lecture 13

Multiplexers and Three-State Gates

The 2-to-1 and 4-to-1 multiplexer · functions from a multiplexer
· expansion · three-state buffers and buses



Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapter 4, §4.11

Today's roadmap

1

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Multiplexers

Select one of many inputs

2

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Functions

Any function, no K-map

3

§4.11

Expansion

Bigger MUXes from smaller

4

§4.11

Three-state

Sharing a bus safely

By the end of class you can...

- Derive a 2-to-1 multiplexer from gates
- Implement an n -variable function on a $2^{(n-1)}$ -to-1 multiplexer
- Choose the residual variable to simplify the wiring
- Build larger multiplexers from smaller ones
- Use three-state buffers on a shared bus without contention

Agenda for today's session

Intro

MUX

Functions

Break

Expand

3-state

Wrap

0:00

0:05

0:25

0:55

1:05

1:20

1:40

1:50

Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome, roadmap	Homework 6 collected
0:05 – 0:25	2-to-1 from gates; the 4-to-1 MUX	Example 1
0:25 – 0:55	Functions on a multiplexer	Examples 2, 3 · Your turn 1
0:55 – 1:05	Break	
1:05 – 1:20	Multiplexer expansion	Example 4
1:20 – 1:40	Three-state buffers and buses	Your turn 2
1:40 – 1:50	Verilog; wrap-up	Lab G preview

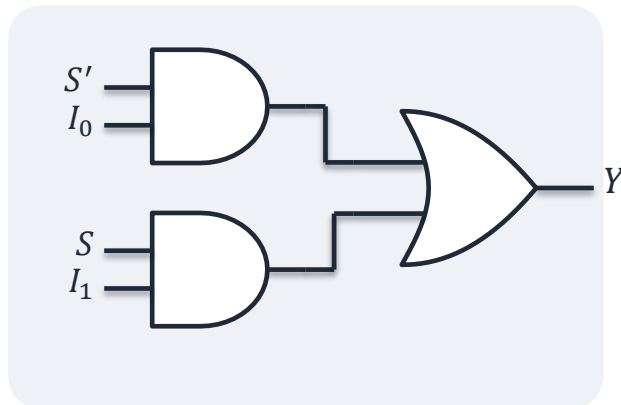
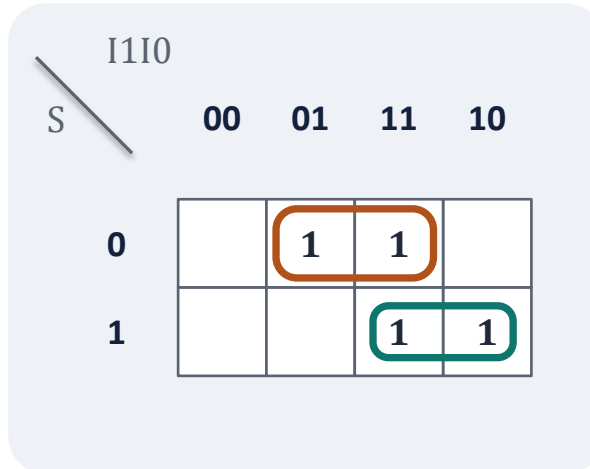
Example 1 — A 2-to-1 MUX from gates

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EXAMPLE

Output $Y = I_0$ when $S = 0$ and $Y = I_1$ when $S = 1$. Derive the circuit.

$S I_1 I_0$	Y
0 0 0	0
0 0 1	1
0 1 0	0
0 1 1	1
1 0 0	0
1 0 1	0
1 1 0	1
1 1 1	1

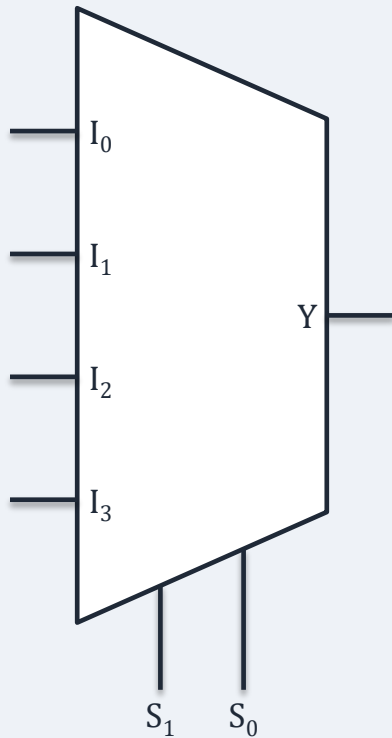


- 1 Rule: $Y = I_0$ when $S = 0$, $Y = I_1$ when $S = 1$
- 2 Treat S as one more input: $Y = \Sigma m(1,3,6,7)$
- 3 Pair $m1$ (001), $m3$ (011): $S = 0, I_0 = 1 \rightarrow S'I_0$
- 4 Pair $m6$ (110), $m7$ (111): $S = 1, I_1 = 1 \rightarrow SI_1$
- 5
$$Y = S'I_0 + SI_1$$
- 6 S enables exactly one AND gate

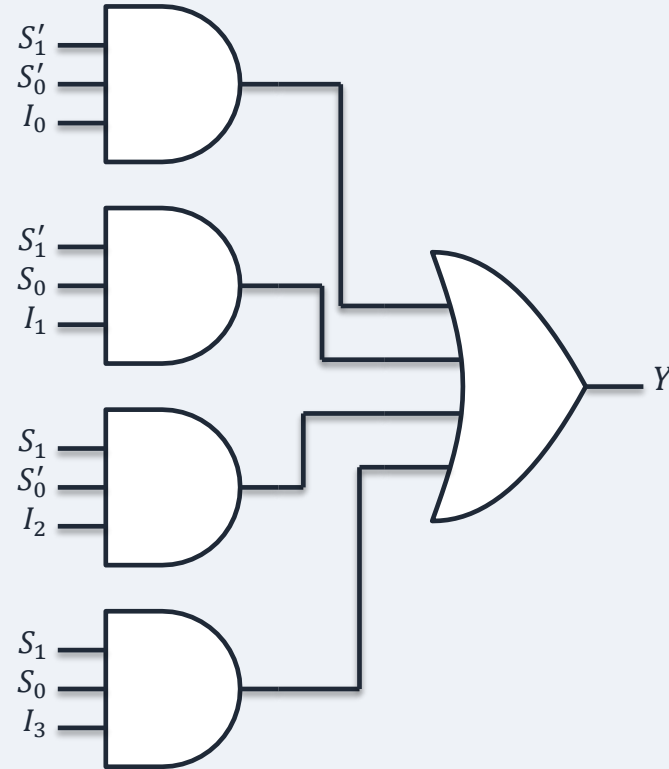
The 4-to-1 multiplexer

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Symbol



Gate level



S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

In general

2^n data inputs, n selects: a decoder whose outputs each gate one input.

Functions from a multiplexer

§4.11

Procedure

An n -variable function fits a 2^{n-1} -to-1 MUX.

Put $n - 1$ variables on the select lines; the last one is the residual variable r .

For each select code, look at the two rows $r = 0$ and $r = 1$.

Wire that data input to 0, 1, r or r' .

F at $r = 0$	F at $r = 1$	Data input
0	0	0 (ground)
1	1	1 (V_{CC})
0	1	r
1	0	r'

No K-map is needed: the multiplexer does the decoding. At most one inverter is added.

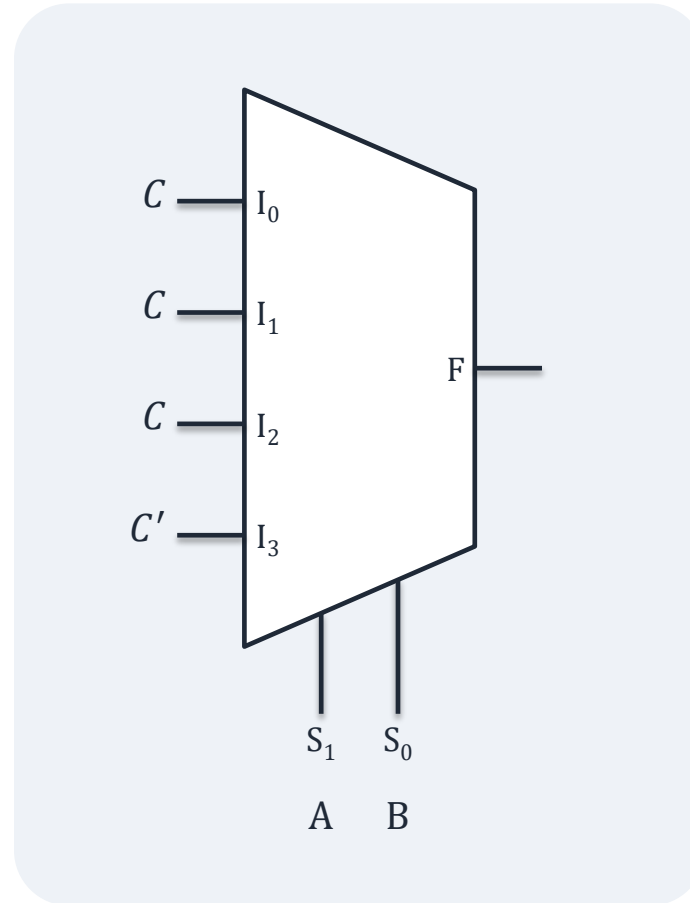
Example 2 — Three variables, 4-to-1

§4.11

EXAMPLE

Implement $F(A, B, C) = \Sigma m(1, 3, 5, 6)$ on a 4-to-1 MUX with A, B on the selects.

A B	C=0	C=1	Input
00	0	1	$I_0 = C$
01	0	1	$I_1 = C$
10	0	1	$I_2 = C$
11	1	0	$I_3 = C'$



- 1 For each AB , read F at $C = 0$ and $C = 1$
- 2 $AB = 00$: $m_0 = 0, m_1 = 1 \rightarrow C$
- 3 $AB = 01$: $m_2 = 0, m_3 = 1 \rightarrow C$
- 4 $AB = 10$: $m_4 = 0, m_5 = 1 \rightarrow C$
- 5 $AB = 11$: $m_6 = 1, m_7 = 0 \rightarrow C'$
- 6 $I_0 = I_1 = I_2 = C, I_3 = C'$
- 7 Check m_6 (110): picks $I_3 = C' = 1$ ✓

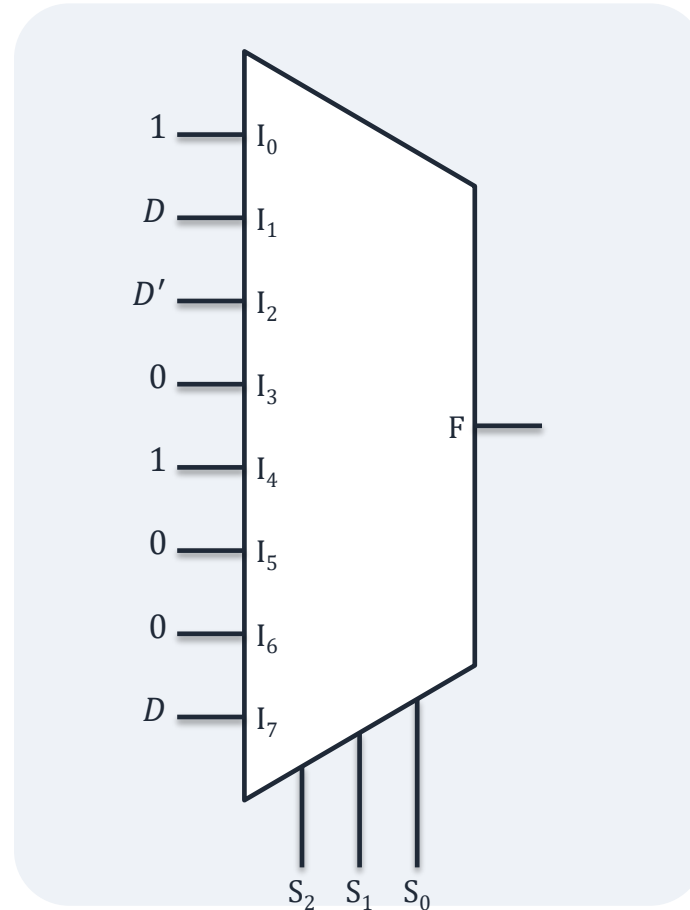
Example 3 — Four variables, 8-to-1

§4.11

EXAMPLE

Implement $F(A, B, C, D) = \Sigma m(0, 1, 3, 4, 8, 9, 15)$ on an 8-to-1 MUX.

A B C	D=0	D=1	Input
000	1	1	$I_0 = 1$
001	0	1	$I_1 = D$
010	1	0	$I_2 = D'$
011	0	0	$I_3 = 0$
100	1	1	$I_4 = 1$
101	0	0	$I_5 = 0$
110	0	0	$I_6 = 0$
111	0	1	$I_7 = D$



- 1 A, B, C on the selects; D residual
- 2 Pair values: $00 \rightarrow 0$; $11 \rightarrow 1$; $01 \rightarrow D$; $10 \rightarrow D'$
- 3 000 : $1, 1 \rightarrow I_0 = 1$; 001 : $0, 1 \rightarrow I_1 = D$
- 4 010 : $1, 0 \rightarrow I_2 = D'$; 011 : $0, 0 \rightarrow I_3 = 0$
- 5 100 : $1, 1 \rightarrow I_4 = 1$; 101 : $0, 0 \rightarrow I_5 = 0$
- 6 110 : $0, 0 \rightarrow I_6 = 0$; 111 : $0, 1 \rightarrow I_7 = D$
- 7 $1 \rightarrow V_{CC}$, $0 \rightarrow \text{ground}$
- 8 Check $m15$: $111 \rightarrow I_7 = D = 1 \checkmark$

Your turn 1 — Choose the residual

§4.11

YOUR TURN

≈ 10 min

$F(A, B, C) = \Sigma m(1, 2, 5, 7)$ on a 4-to-1 multiplexer.

(a) Put A, B on the selects (C residual). Find I_0 – I_3 .

(b) Put B, C on the selects (A residual). Find I_0 – I_3 .

(c) Which wiring needs no inverter on the residual? Which is cheaper overall?

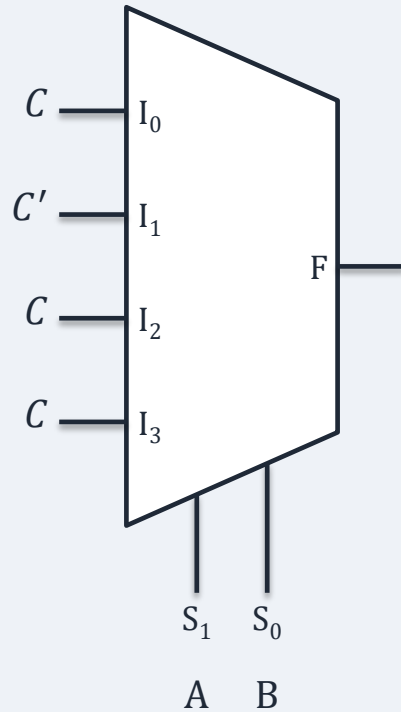
In (b), pair the minterms by their BC value.

Your turn 1 — Solution

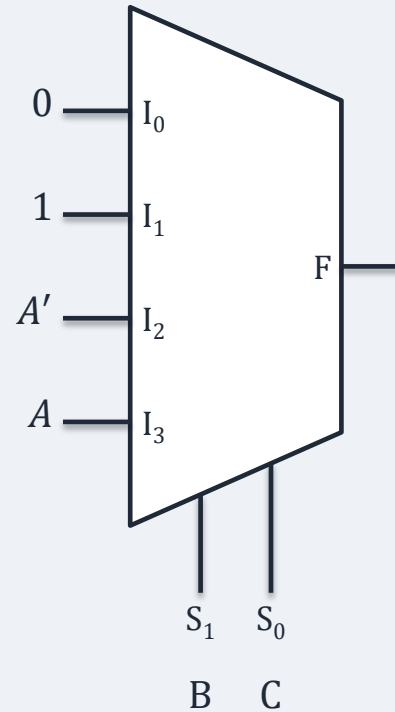
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SOLUTION

(a) A, B select



(b) B, C select



- 1 (a) $AB = 00$: $m_0 = 0$, $m_1 = 1 \rightarrow C$
- 2 $AB = 01$: $m_2 = 1$, $m_3 = 0 \rightarrow C'$
- 3 $AB = 10$: $0, 1 \rightarrow C$; $AB = 11$: $0, 1 \rightarrow C$
- 4 (b) For each BC : F at $A = 0$, then $A = 1$
- 5 $BC = 00$: $m_0, m_4 = 0$, $0 \rightarrow 0$
- 6 $BC = 01$: $m_1, m_5 = 1$, $1 \rightarrow 1$
- 7 $BC = 10$: $m_2, m_6 = 1$, $0 \rightarrow A'$
- 8 $BC = 11$: $m_3, m_7 = 0$, $1 \rightarrow A$
- 9 (c) Both need one inverter; (b) needs A on only 2 inputs

Break

10 minutes — we resume at the 1:05 mark

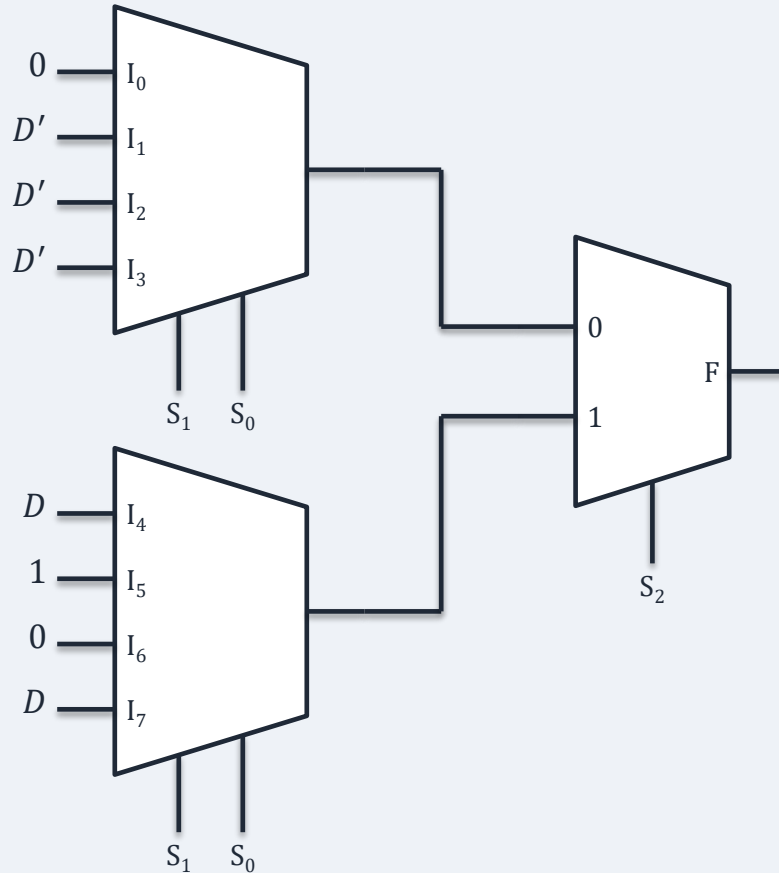
Next: multiplexer expansion and three-state buffers

Example 4 — Multiplexer expansion

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EXAMPLE

Build an 8-to-1 MUX from 4-to-1 MUXes, then use it for $F = \Sigma m(2,4,6,9,10,11,15)$.

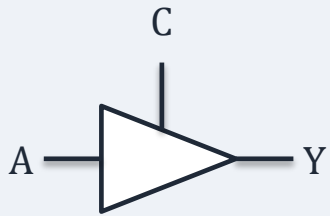


- 1 $S_1 S_0$ go to both 4-to-1s; the 2-to-1 on S_2 picks one
- 2 $A, B, C \rightarrow S_2, S_1, S_0$; D residual
- 3 Pairs ($D = 0, D = 1$): 000: 0, 0 $\rightarrow$ 0; 001: 1, 0 $\rightarrow D'$
- 4 010: 1, 0 $\rightarrow D'$; 011: 1, 0 $\rightarrow D'$
- 5 100: 0, 1 $\rightarrow D$; 101: 1, 1 $\rightarrow$ 1
- 6 110: 0, 0 $\rightarrow$ 0; 111: 0, 1 $\rightarrow D$
- 7 Check $ABC = 110$: lower MUX, input 2: $I_6 = 0$ ✓
- 8 Alternative: enables S'_2 and S_2 , then OR

Three-state buffers

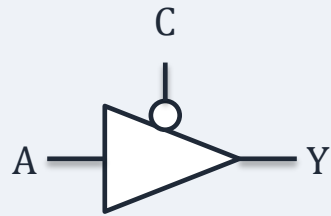
§4.11

Active-high control



C	Y
0	Hi-Z
1	A

Active-low control

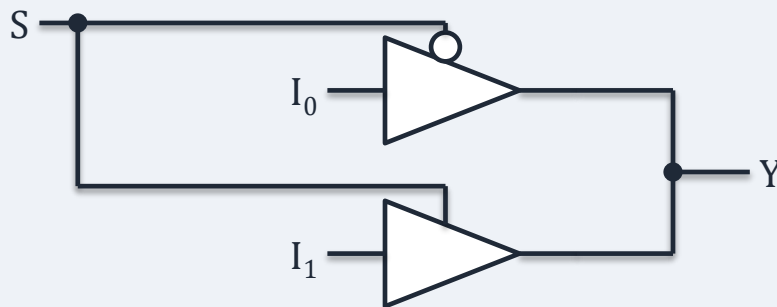


C	Y
1	Hi-Z
0	A

The third state

High impedance (Hi-Z): the output is disconnected, neither 0 nor 1.

Many Hi-Z outputs can share one wire; the one enabled drives it.



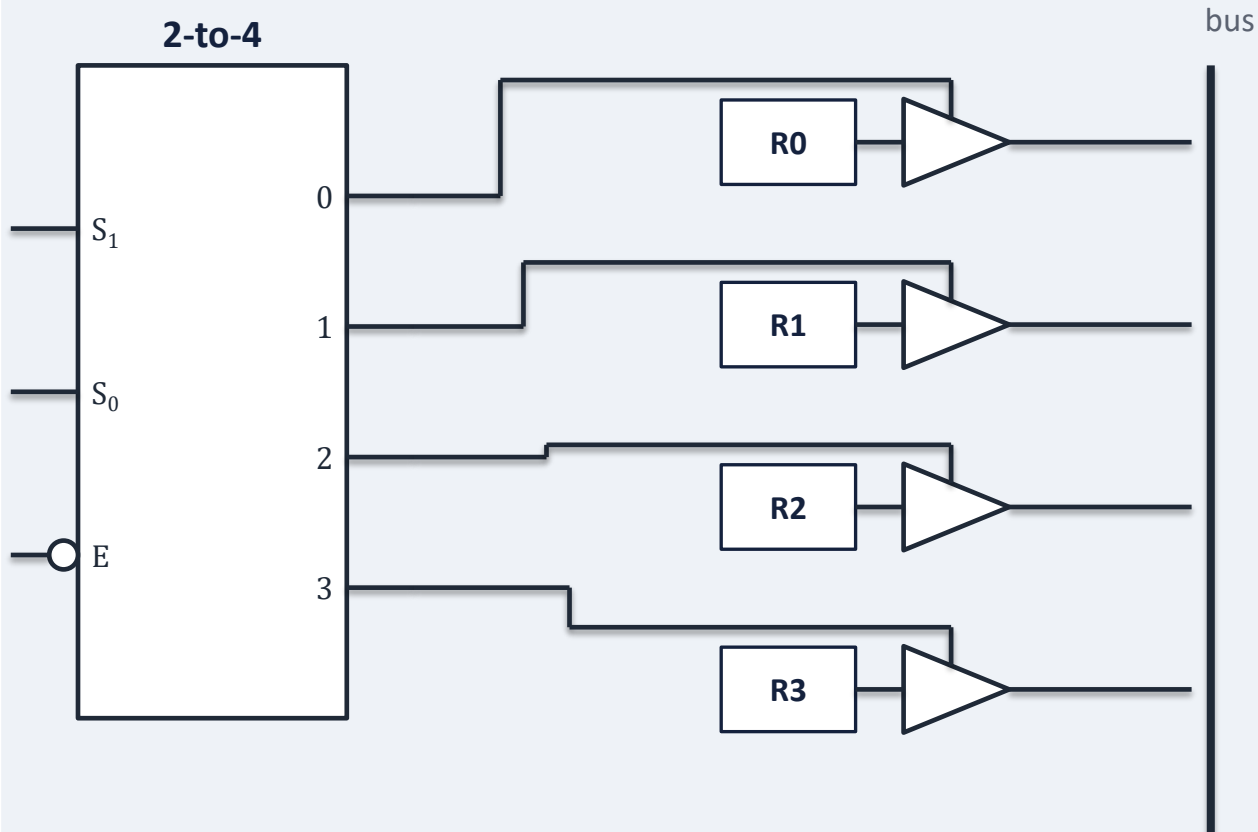
A 2-to-1 MUX from two buffers

$S = 0$ enables the top buffer, $S = 1$ the bottom one: exactly one drives Y .

Buses and contention

§4.11

Four registers share one bus line



The rule

Exactly one driver enabled at a time.

A decoder makes that automatic: only one output is ever 1.

$S_1S_0 = 10$: R_2 drives the bus; the others are Hi-Z.

Decoder disabled: nobody drives; the bus is free for another device.

An 8-bit bus uses eight buffers per register, one wire per bit.

Your turn 2 — Three-state logic

§4.11

YOUR TURN

≈ 10 min

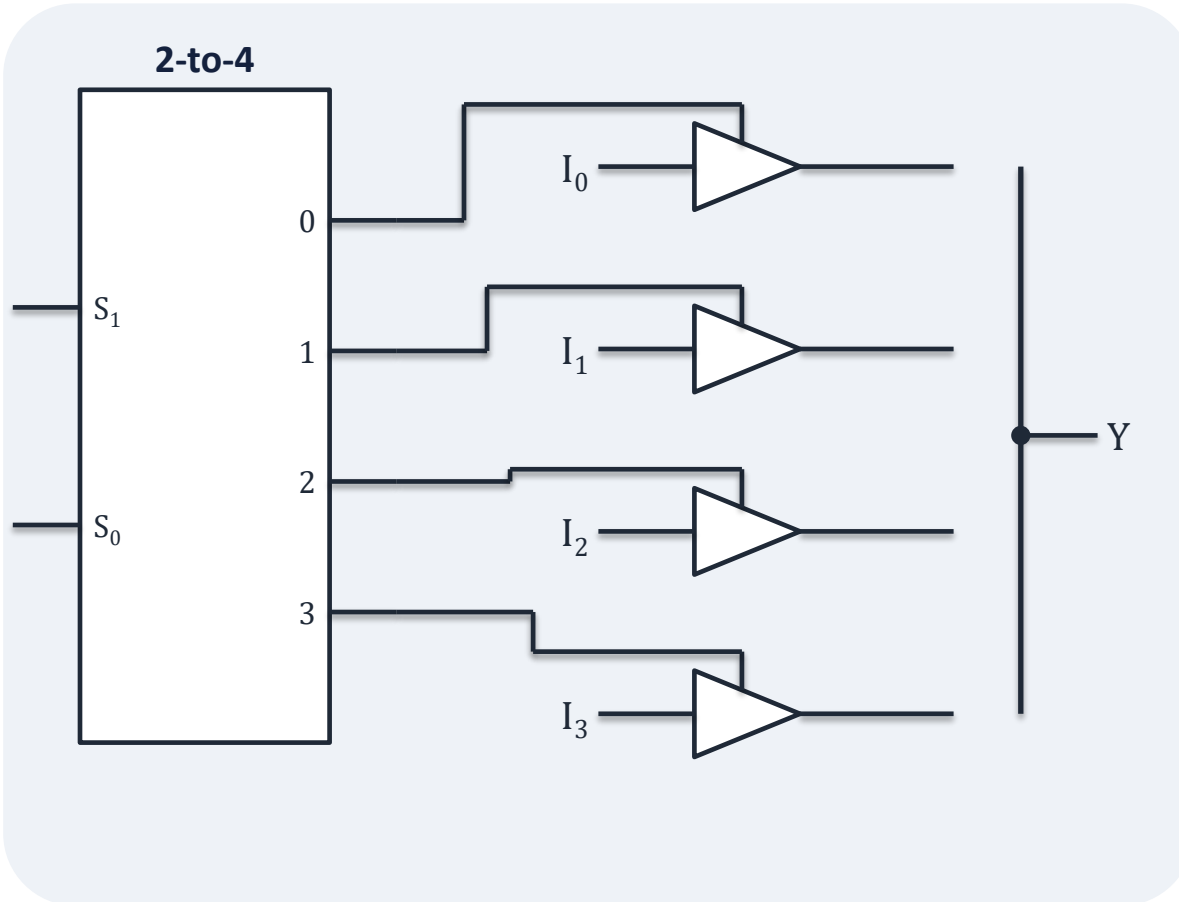
- (a) Two enabled three-state outputs drive the same wire, one HIGH and one LOW. What happens?
- (b) Build a 4-to-1 multiplexer from three-state buffers and a 2-to-4 decoder. Sketch it.
- (c) In the bus of the previous slide, which register drives the bus for $S_1S_0 = 01$? What if the decoder is disabled?

Think of each buffer as a switch.

Your turn 2 — Solution

§4.11

SOLUTION



- 1 (a) One output pulls up, the other down: **bus contention**
- 2 The level is undefined and a large current flows
- 3 (b) The 2-to-4 decoder turns S_1S_0 into one enable line
- 4 Output k enables only the buffer carrying I_k
- 5 Join the four outputs: $Y = I_k$ when $S_1S_0 = k$
- 6 (c) $S_1S_0 = 01$: decoder output 1 $\rightarrow R_1$ drives the bus
- 7 Decoder disabled: all Hi-Z; the bus floats (free)

Multiplexers in Verilog

§4.12

Three descriptions

```
// 2-to-1, conditional operator
assign Y = S ? I1 : I0;

// 4-to-1, behavioral
always @(*)
  case (S)
    2'b00: Y = I0;    2'b01: Y = I1;
    2'b10: Y = I2;    default: Y = I3;
  endcase

// three-state buffer
assign Y = En ? A : 1'bz;
```

Notes

``?:`` is the Verilog form of $Y = S'I_0 + SI_1$.

In the case form, Y must be declared reg; default avoids an unintended latch.

1'bz is the Hi-Z value: a synthesizer infers a three-state buffer.

Lab G builds the 4-to-1 MUX from gates, then checks it against the case version.

Concept check

1

How many select lines does a 16-to-1 multiplexer need?

2

What size MUX implements any 5-variable function?

3

What are the four possible values of a data input in the MUX method?

4

Why use a decoder to drive three-state enables on a bus?

Key takeaways

2-to-1

$$Y = S'I_0 + SI_1$$

2^n -to-1

n select lines

Functions

n vars on 2^{n-1} -to-1

Data inputs

$0, 1, r, r'$

Three-state

0, 1 or Hi-Z

Bus rule

one driver at a time

A multiplexer is a programmable truth table.

Homework #7 due at the start of Lecture 16

Next: binary adders and subtractors